Python UNV Converter Short Guide

Read Original Article by Dip28 here:
http://dip28p.web.fc2.com/calculix/gimpconvert/index.htm
(or in PDF copy in the folder: \c4w\Hlp\conv\Unv_by_Dip28.pdf).

GMSH saving options are shown on Fig.1:

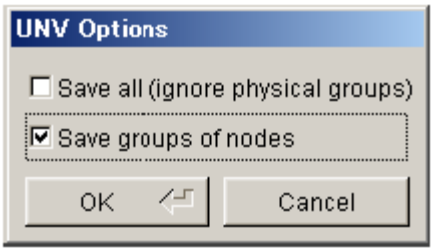


Fig. 1 Saving .UNV file in GMSH

Short Description:

1. Create the geometrical model in GMSH
2. Mark physical surfaces and volumes to get the groups of nodes as the basis for future FE model.
3. Save as .UNV with "groups of nodes" option (Fig. 1)
4. Run **CalculiXForWin** and click on Tree Menu > Conversion > .UNV (Fig. 2).
5. Select path to .UNV file (Fig. 3) and click on "Convert It" Button. Wait until conversion is done (the DOS window will appear and disappear) Then close "UNV - Conversion" window manually to continue. New **FromUNV_Py.inp.inp** mesh file appears in the working folder.
6. Go to File > Open INP..., set path to "FromUNV_Py.inp.inp" file and re-run CGX.

Also see the detailed example below (Fig.4...and so on).

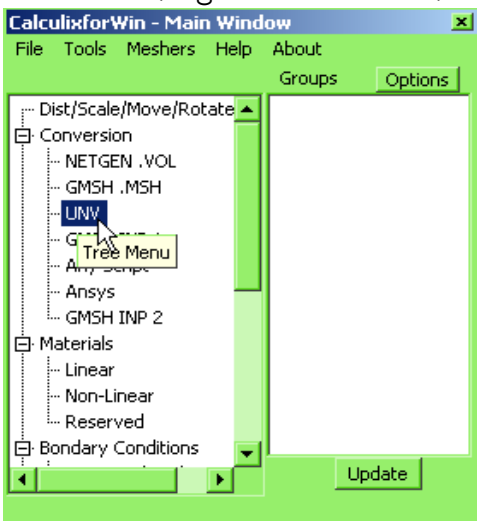


Fig. 2 - Opening of .UNV conversion modal window

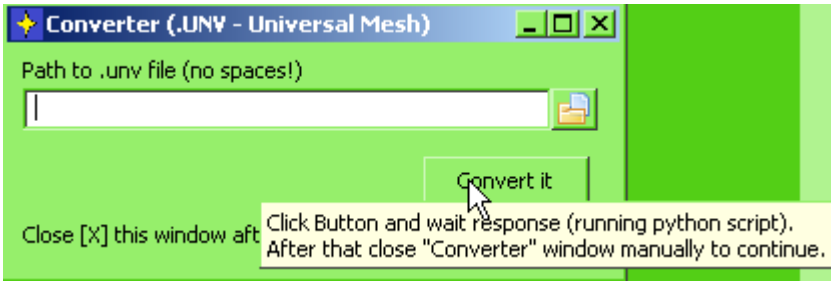


Fig.3 - Conversion Window

Let's look at the example below.

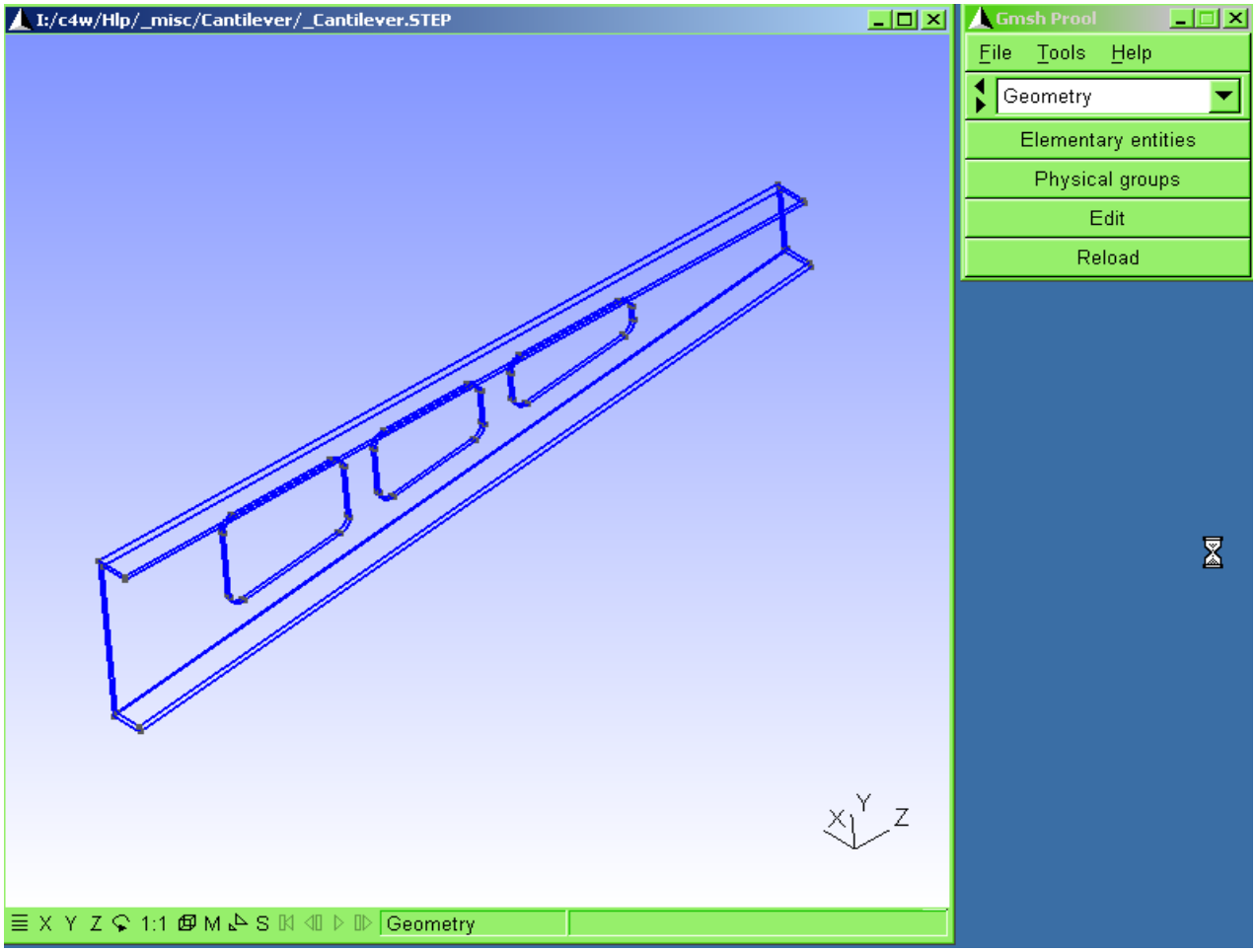


Fig. 4 - Run GMSH (Prool's build) and open _Cantilever.STEP file with geometry in STEP format (see Hlp_misc\Cantilever folder)

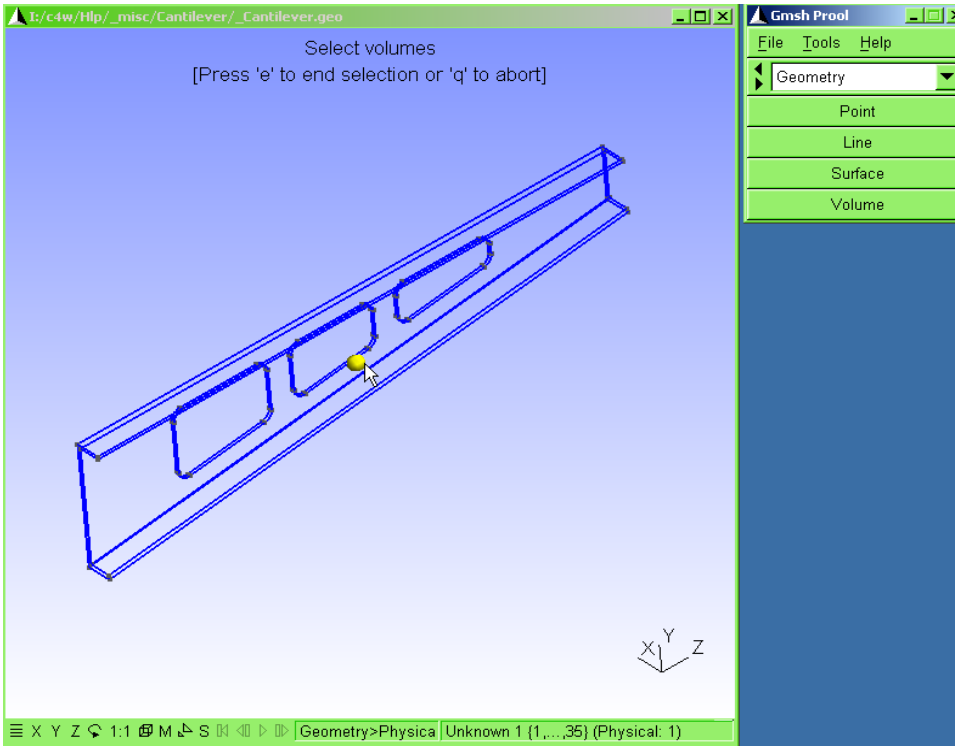


Fig. 5 In GMSH: Geometry>Physical Groups>Add>Volume, click on center of body and type "e" to end selection (see the tips at the top of the graphics window)

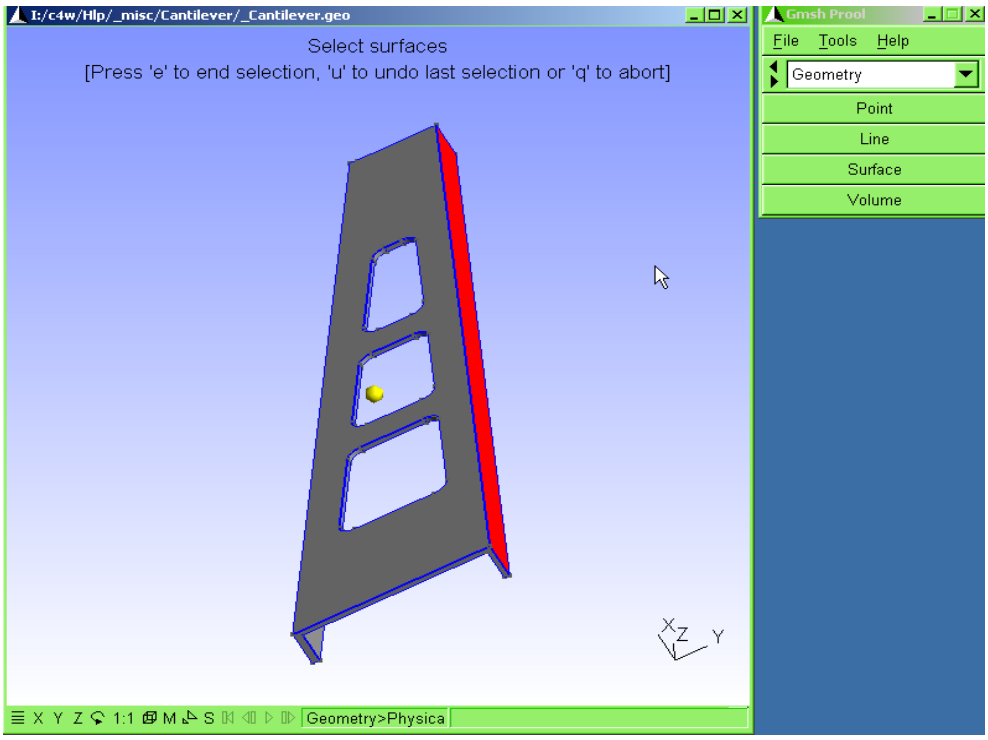


Fig. 6 - In GMSH: Geometry>Physical Groups>Add>Surface and select all surfaces which are planned for applying of boundary conditions (loads and supports). Press "e" to end selection (tips on top).

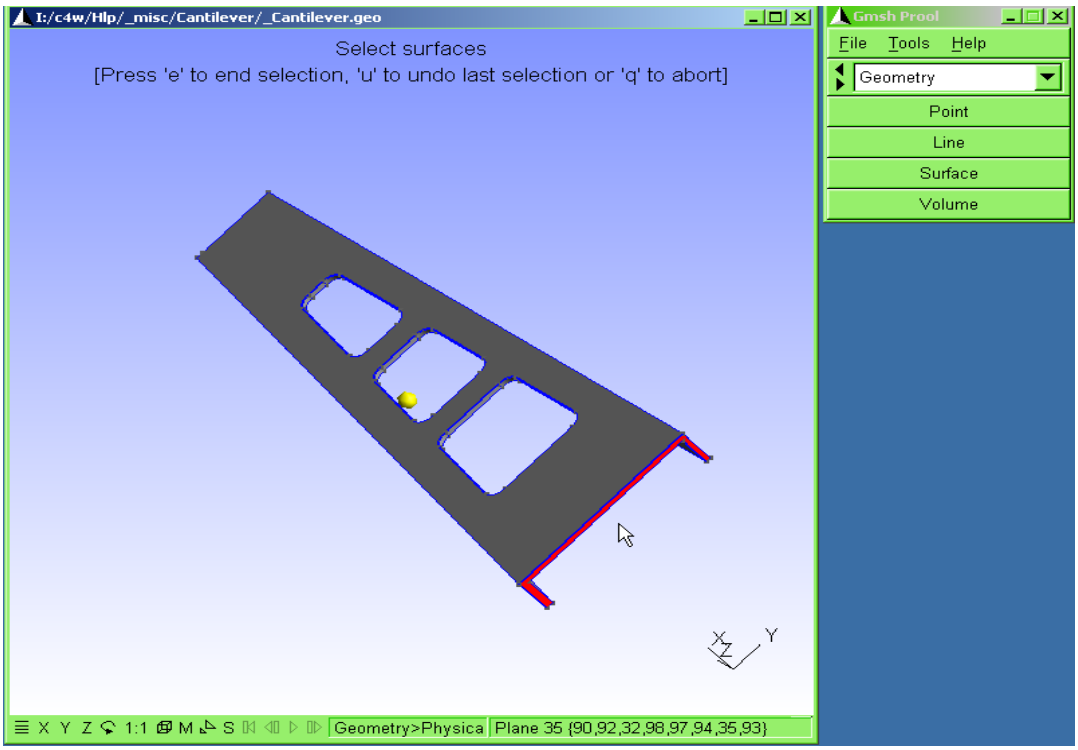


Fig. 7 - see Fig.6 description

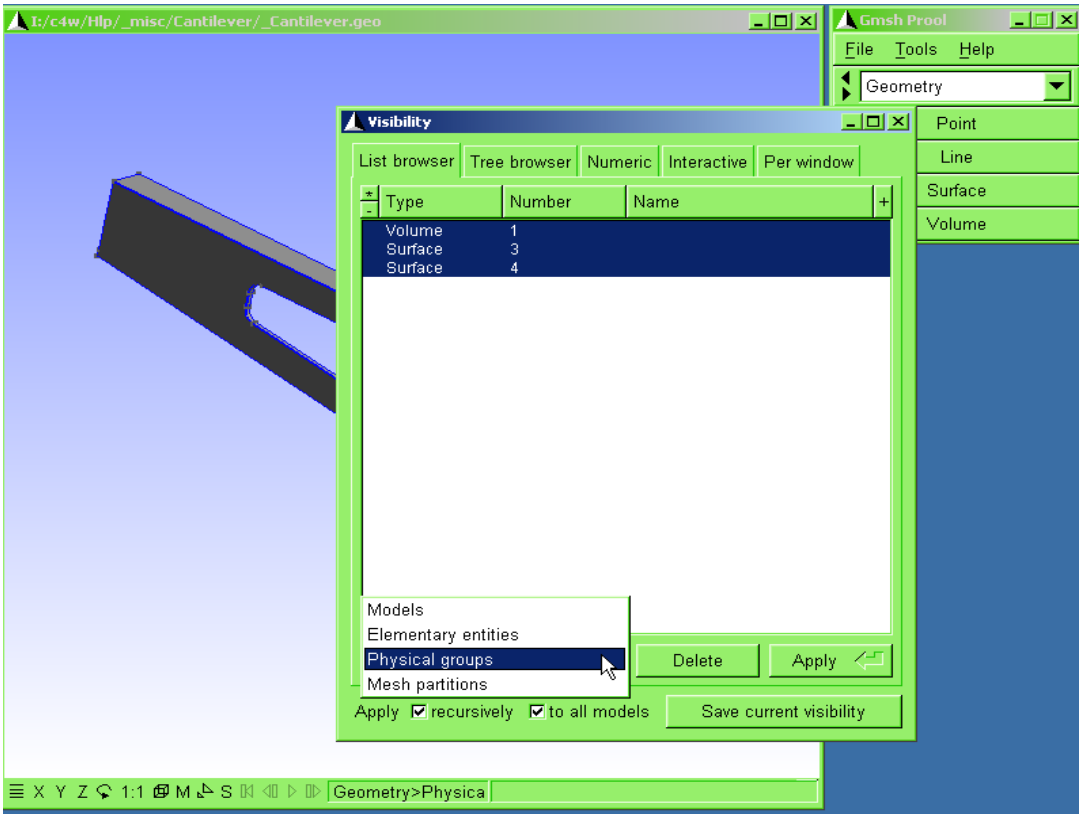


Fig. 8 Go to GMSH Menu>Tools>Visibility and choose Physical Groups in the combobox (to check groups were applied)

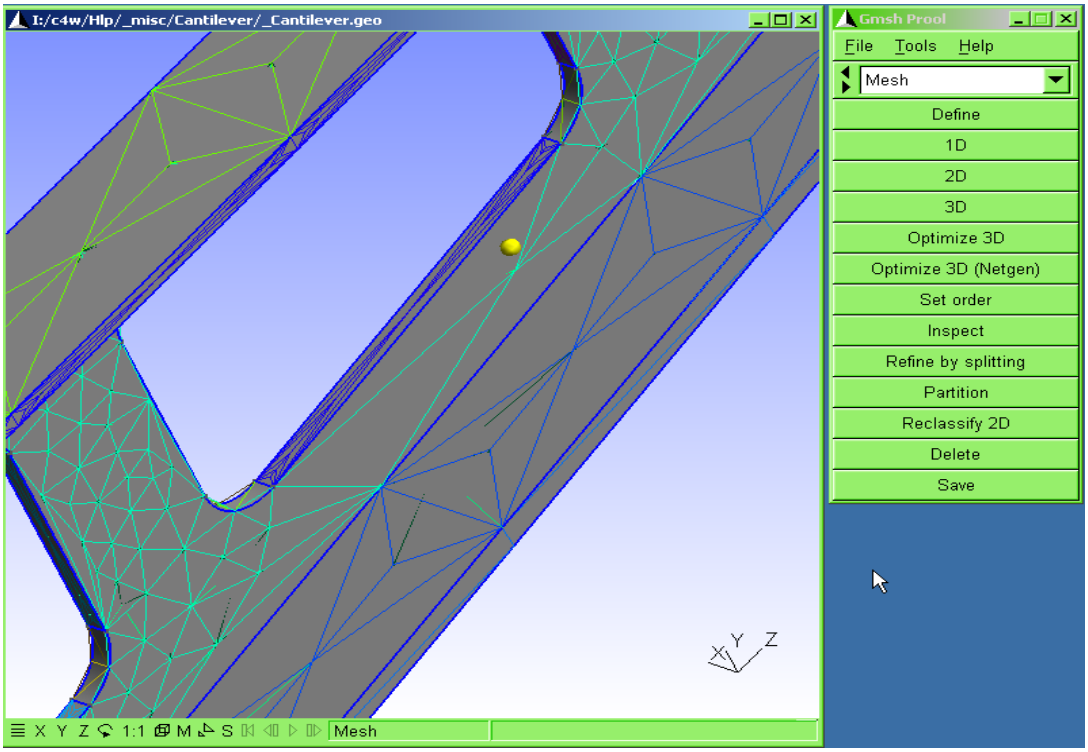


Fig. 9 Go to GMSH>Mesh and press "3D" button (use tree menu in newer GMSH versions). After some time you can see the mesh cover the body. This is terrible mesh with extended elements (often happens in GMSH with the default options). You can't use it for calculation

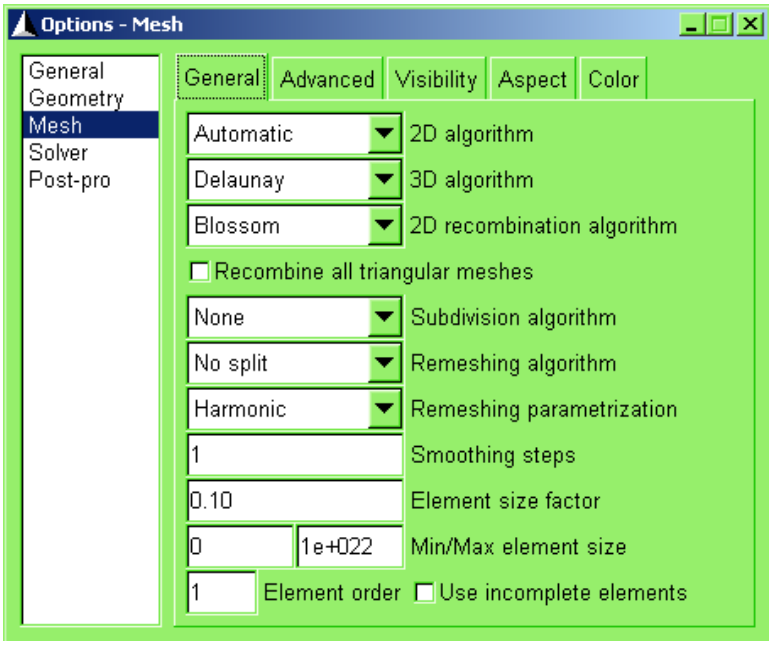


Fig. 10 - Go to GMSH Menu>Tools>Options>Mesh (Listbox)>General (tab) Change element size factor to 0.10

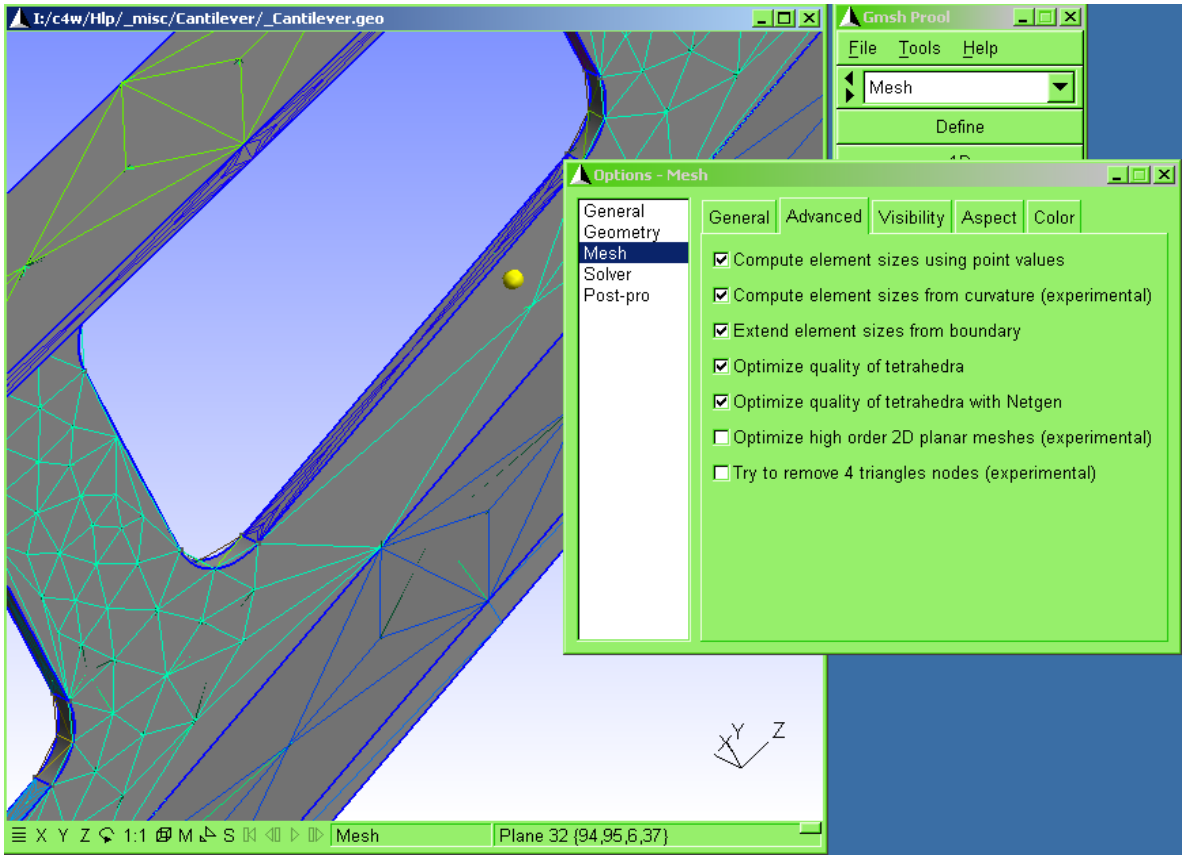


Fig. 11 Check On additional options at Advanced tab.

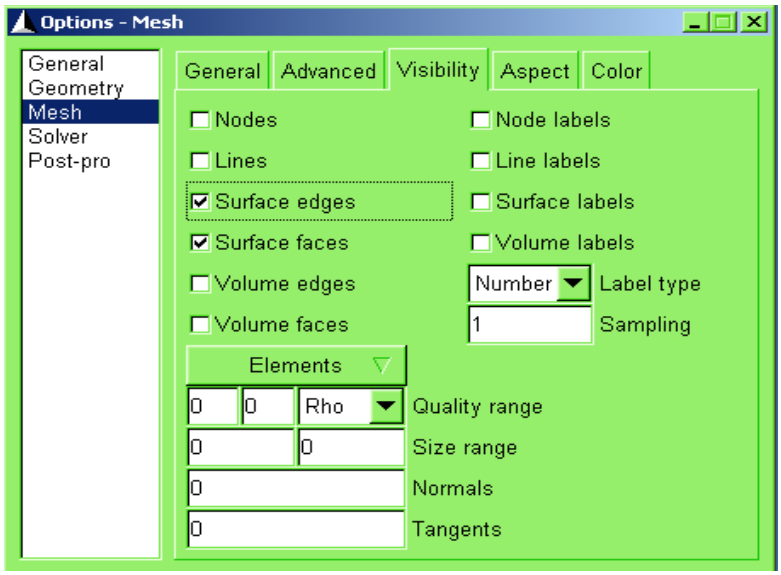


Fig. 12 Check On additional options at Visibility tab (to show the mesh more clear).

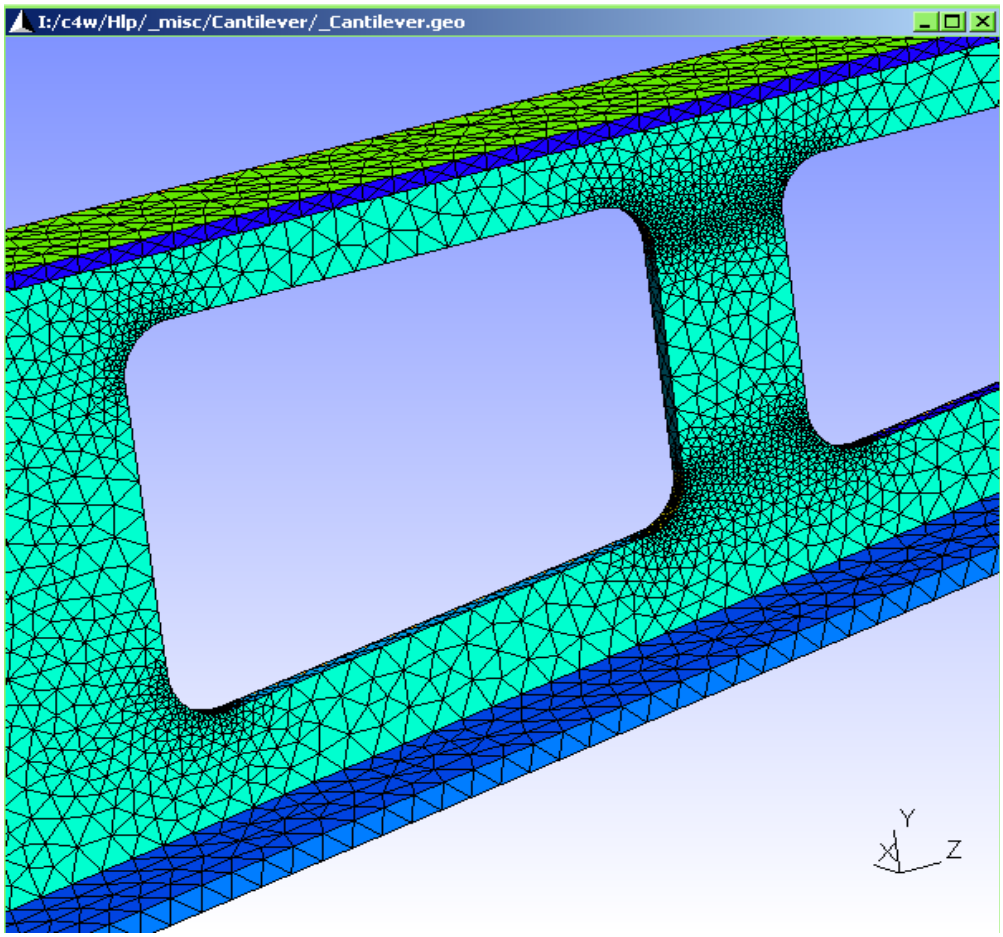


Fig. 13 - Go to GMSH>Mesh and press "3D" button again. Now you can see more or less normal picture without extended elements

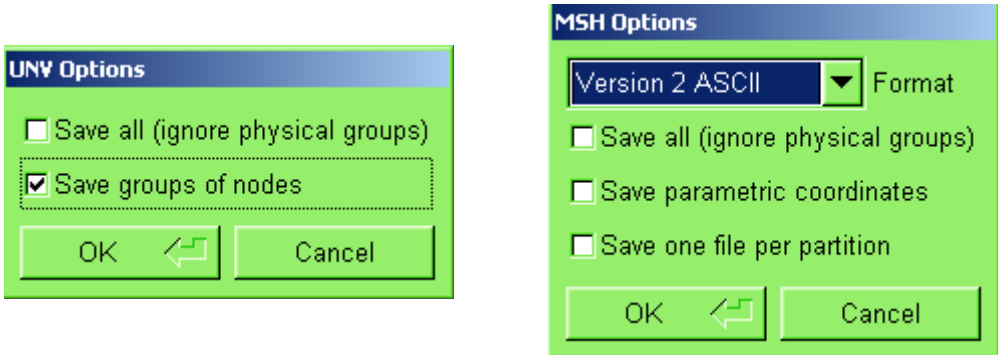


Fig. 14 - GMSH Menu > File > Save As... Select Universal Mesh extension and "Save groups of nodes" option. Then press "OK" and wait.
It is also recommended to save mesh in MESH2.0 format (in this case you will be able to open it again and change if it's needed)

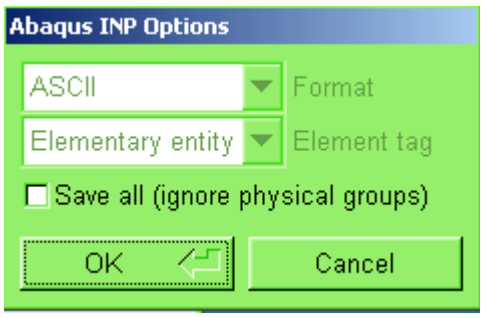


Fig. 15 Prool's special build of GMSH also supports .INP files compatible with CalculiX (in this case you don't need any converters, but you will save all body faces and edges with this option)

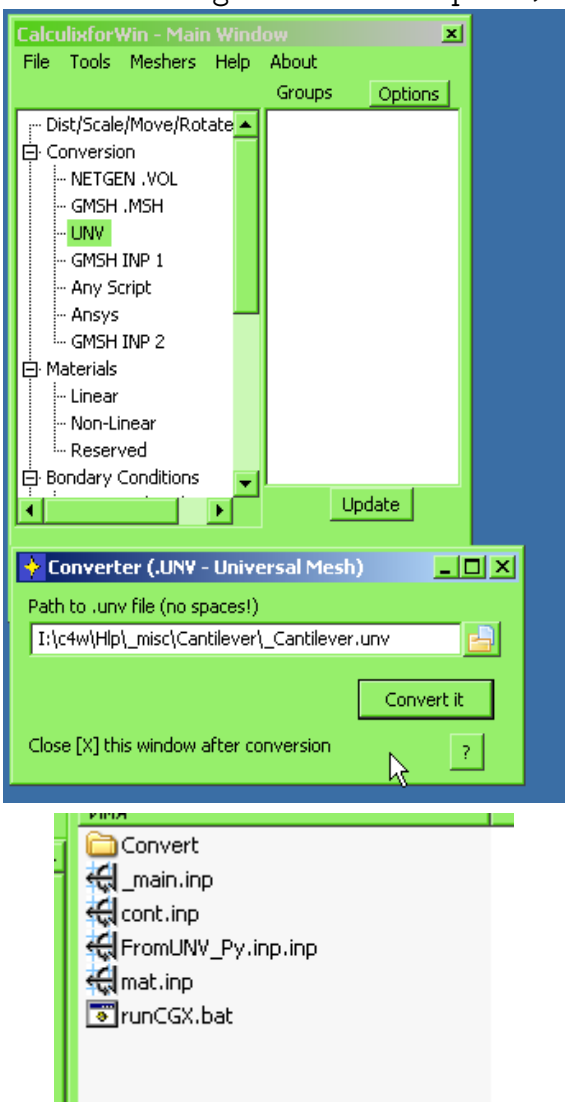


Fig. 16 Go to CalculiXForWin and open UNV conversion window (Fig. 2). Set path to .UNV file (no spaces or cyrillic letters in path!) and press "Convert It". After conversion close the window manually. Now if all is OK you can see FromUNV_Py.inp.inp file in the working folder

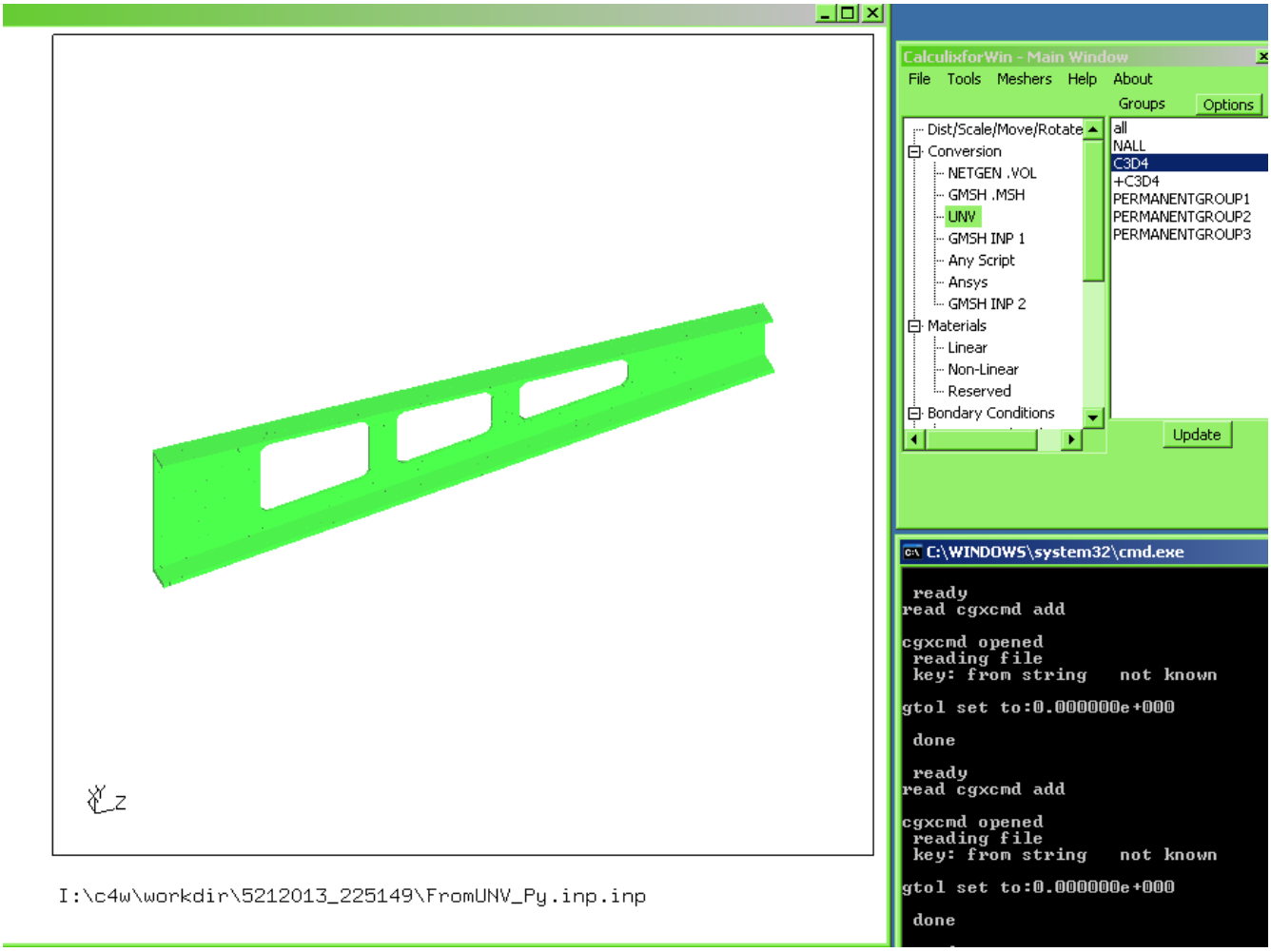


Fig. 17 CalculiXForWin Menu > File> Open INP. Select FromUNV_Py.inp.inp file in the working folder and open it. New CGX window will appear. Press "Update" button under the listbox to see the groups of model (also you can use "prnt se" card in CGX to print list of existing groups)

Current type of element is C3D4. Also three groups of nodes available after conversion (PERMANENTGROUP1, PERMANENTGROUP2 and PERMANENTGROUP3). DoubleClick on it to show. The material was applied by default as steel with imperial units.

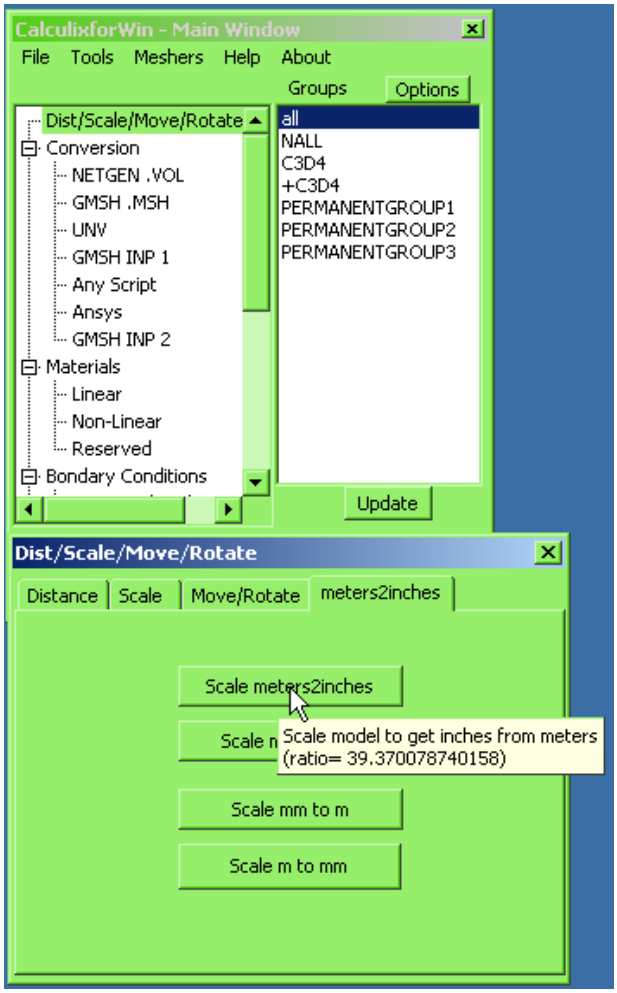


Fig. 18 - It is "number one" deal when you work with mesh, which came from Netgen, GMSH or Salome through STEP format (most convenient for bodies and shells). STEP files contain certain units ("inch", "mm", "m") and current versions of meshers don't support units by themselves and they work with STEP via open-cascade library. According to certain settings in open-cascade source files all units will be transfered into "mm" or "m" (depending on version of build).
It is not problem in general and you just have to know about that and scale your model back depending on unit system you use. If it's SI you have to scale model to get meters. If Imperial - you should get "inches" finally. CalculiXForWin has special window "Dist/Scale/Move/Rotate" to help you scale the model properly. Use "qdis" CalculiX CGX card to be completely sure that your model is correct (it has the same length in certain units you are planning to use).

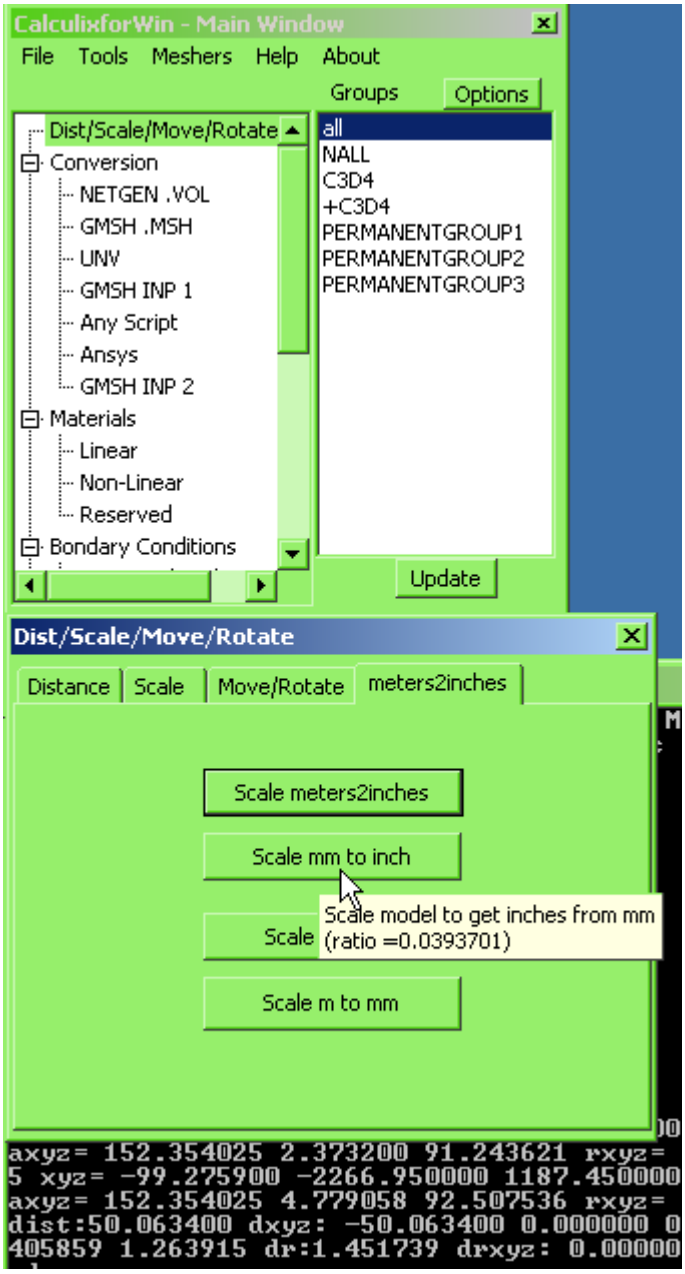


Fig. 19 - In current example the STEP file has "inches" in it and the width of beam is about 2''. In result of measurement with qdis card we obtain dist=50 what supposed to be "mm". Use "scale mm to inch" button to scale the model with ratio = 0.0393 in/mm.

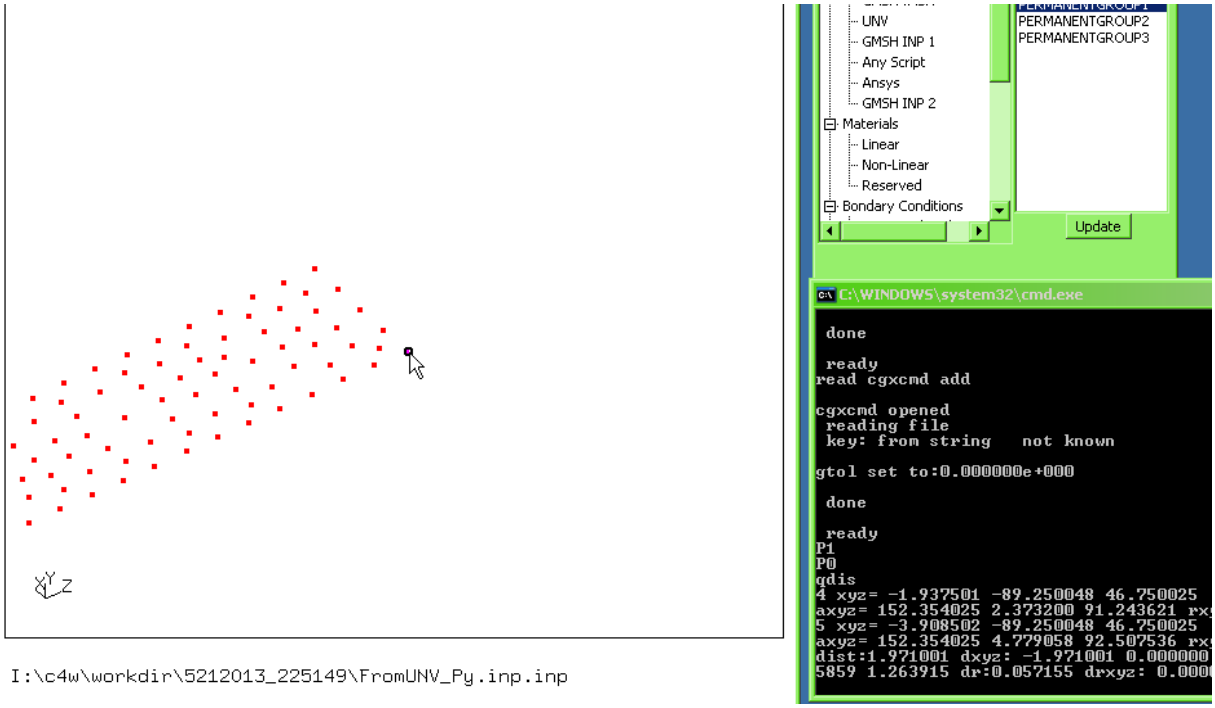


Fig. 20 - Check dimensions with qdis CGX card again. Dist = 2'' now what supposed to be correct.

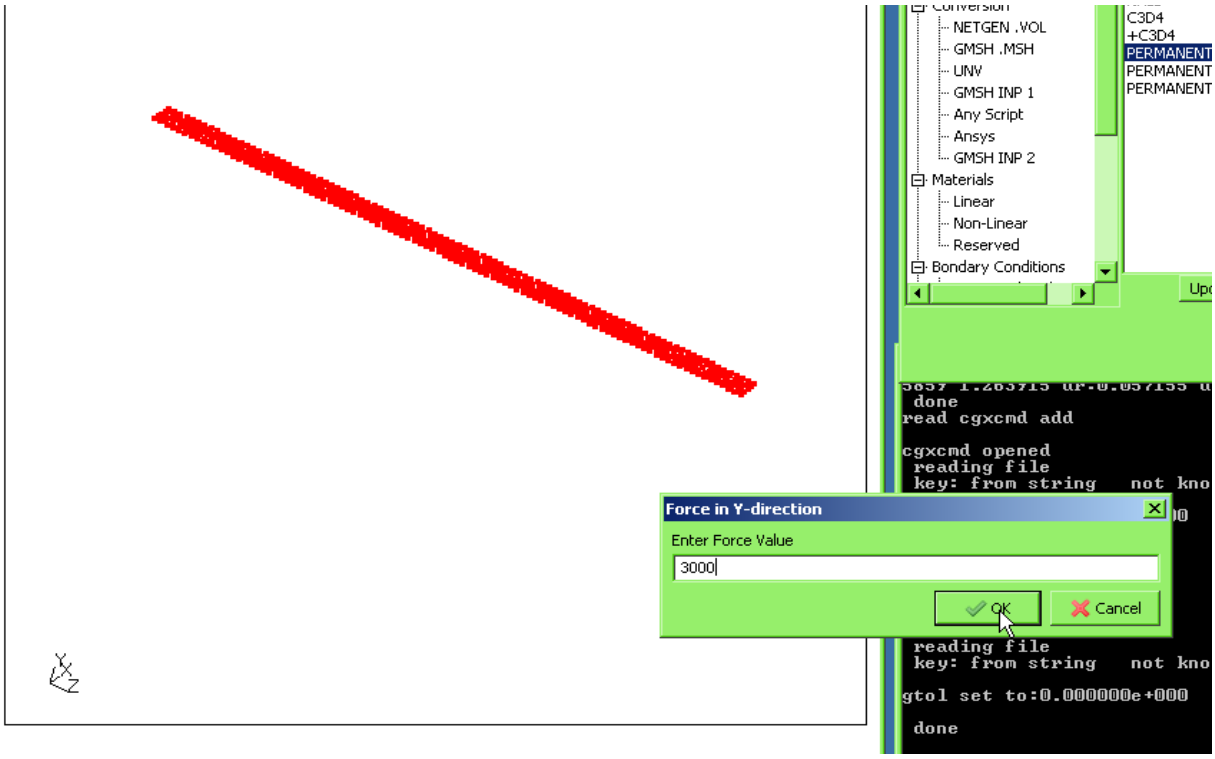


Fig. 21 - Select the group of nodes and apply the load 3000 lb in Y direction using context menu in the list of groups (right mouse button click ...)

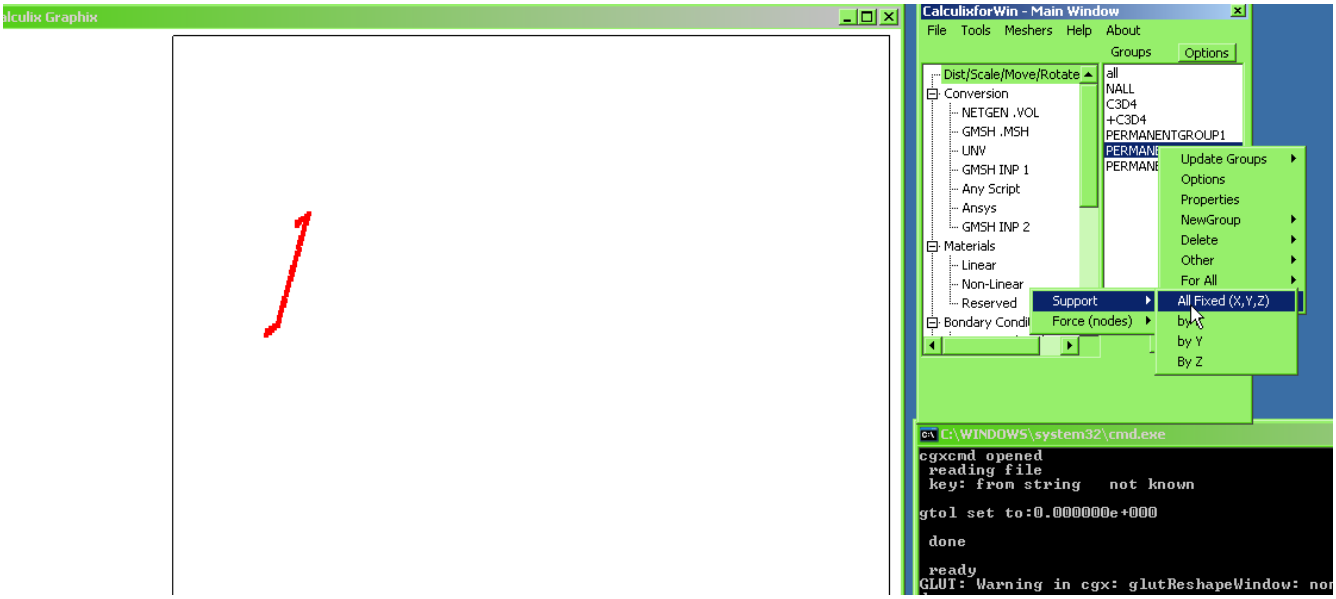


Fig. 22 - Select the group of nodes and apply "all fixed" boundary conditions using context menu in the list of groups (right mouse button click …)

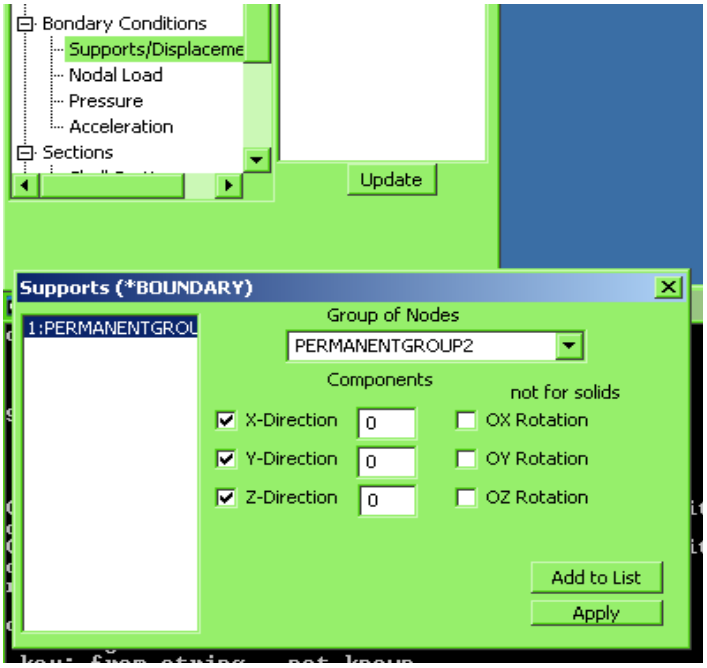


Fig. 23 - Go to the Tree menu > Boundary Conditions > Supports/Displacements and check the BC applied (all fixed means zero displacements of nodes by X, Y, Z direction in Cartesian system).

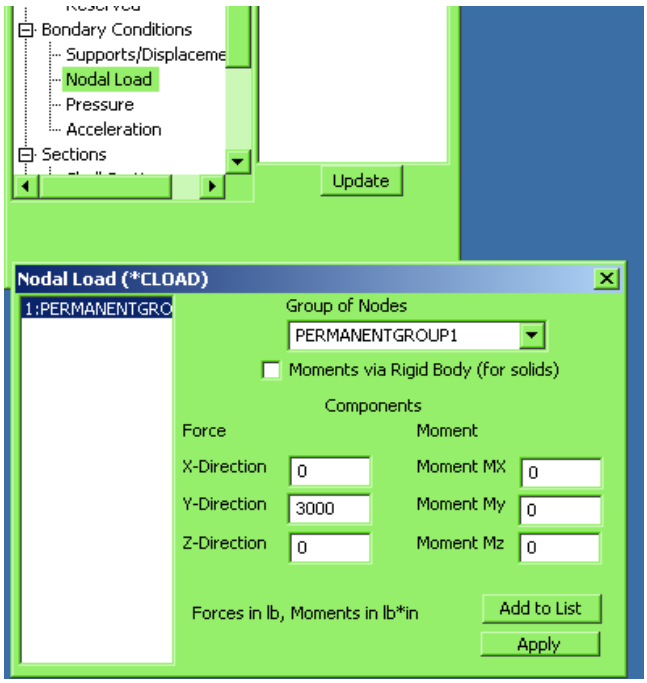


Fig. 24 - Go to the Tree menu > Boundary Conditions > Nodal Load and check the load applied to nodal group ***(Note: you can't use this feature for S6 shell elements! Apply loads to S3 shells and then add middle nodes to S3 to get S6. It is due to mistake in current CCX build and it will be improved in future versions)***

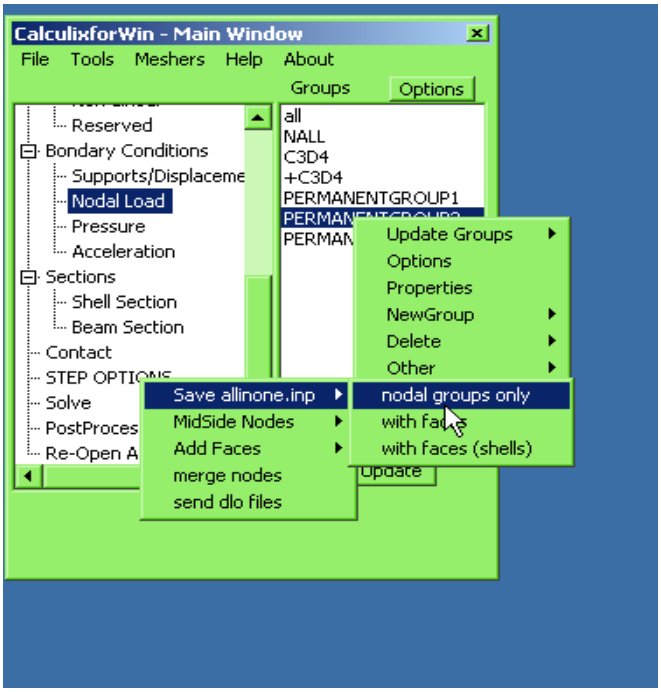


Fig. 25 - Now when the BC's has been applied you have to save allinone.inp file with mesh and groups (you can choose the option "nodal groups only" because we didn't use faces in this example)

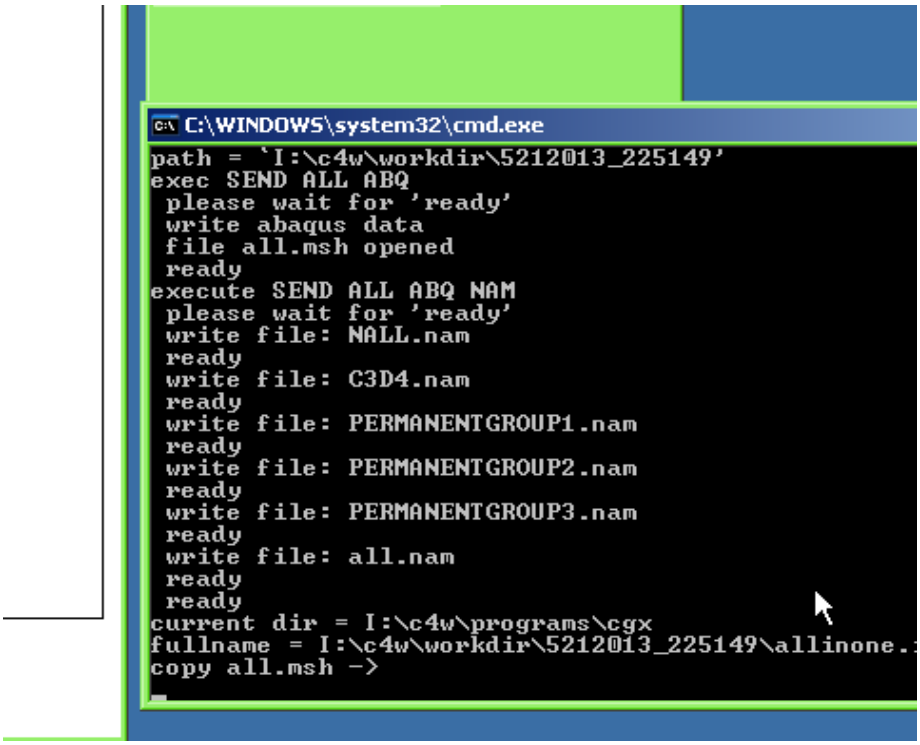


Fig. 26 The procedure of saving allinone.inp file can take long time. Watch for CGX output window (when done/ready) to be sure that it's done before you will start calculation.

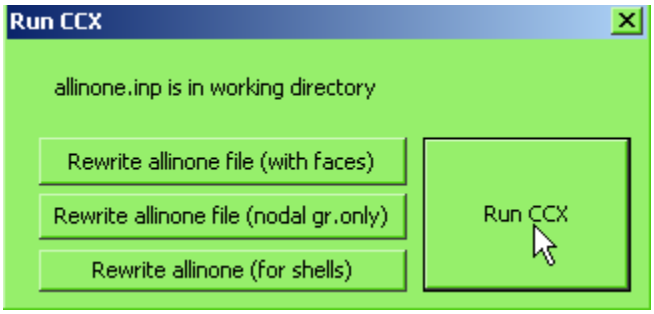


Fig. 27 Go to CalculiXforWin tree menu > Solve and open "Run CCX" window. Allinone file is in working directory and we can start calculation now.

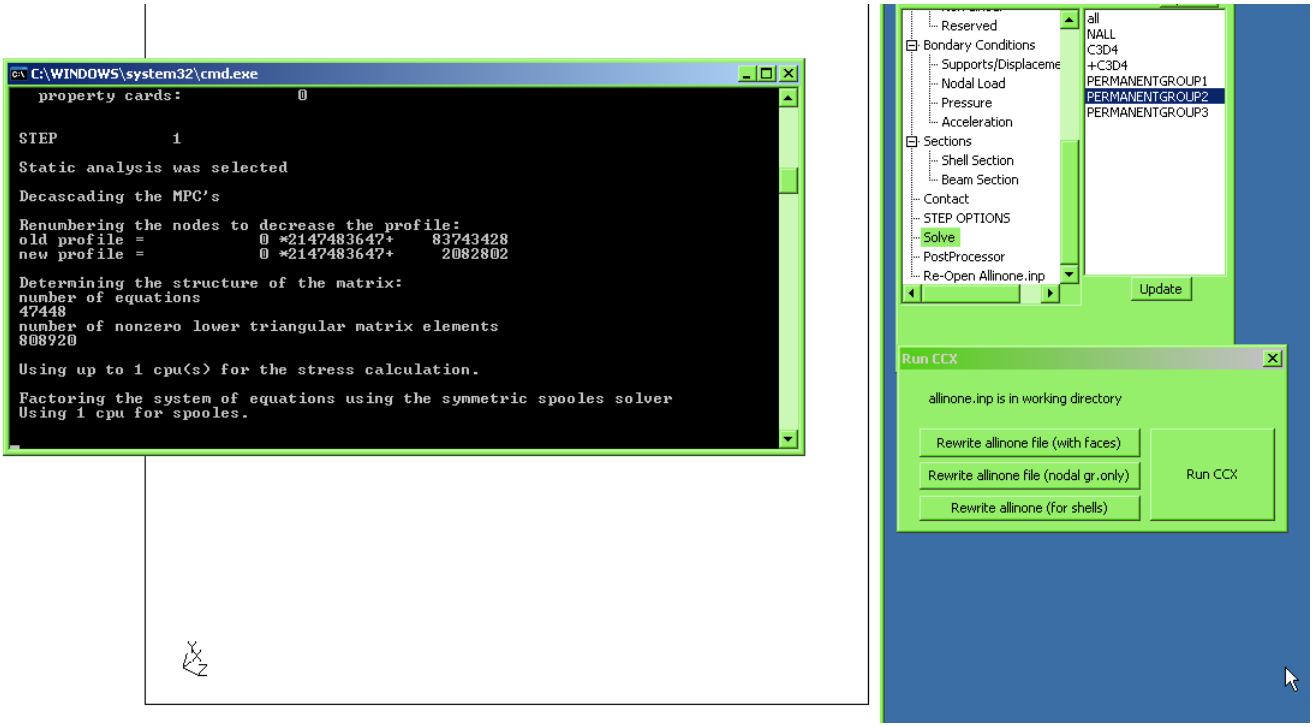


Fig. 28 - Press "Run CCX" button and watch CCX window (calculation is running now). If calculation can't be done due to some mistakes you can go to the working directory and open all .inp files to check data in it.

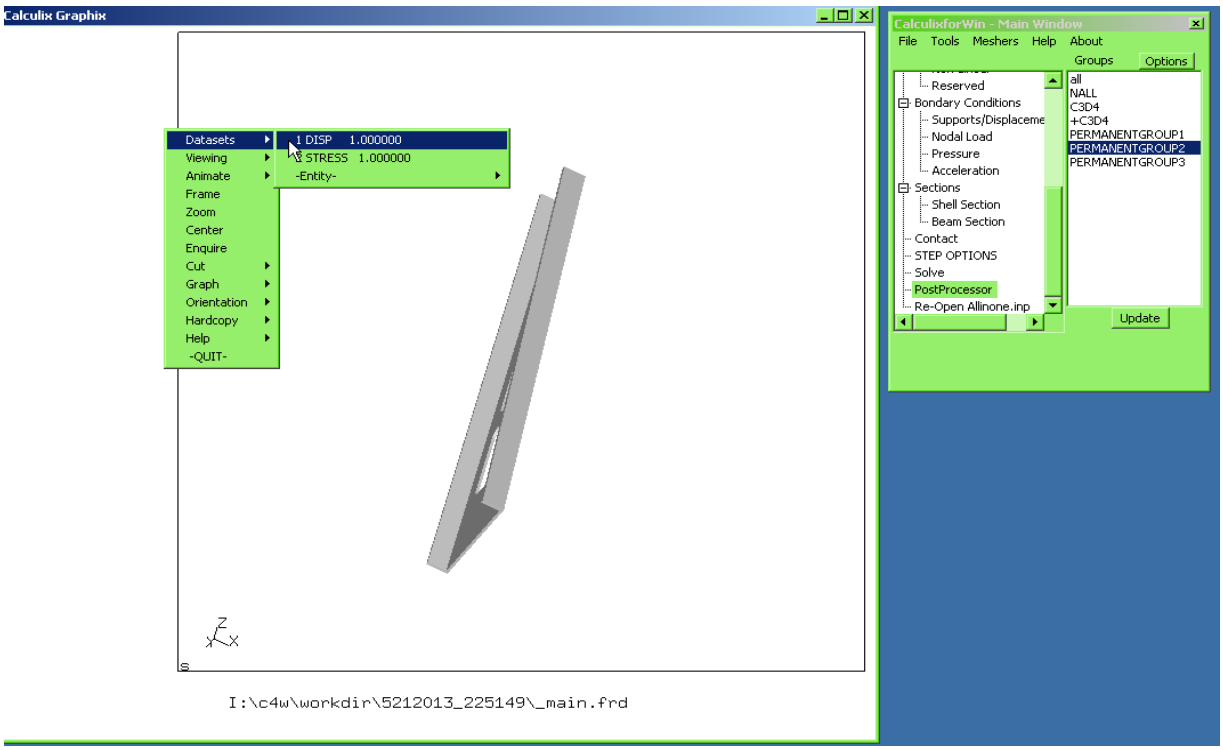


Fig. 29 - In this case there is no problem with calculation and solution has been done. Go to CalculiXForWin Tree menu > Post-processor and run _main.frd file with results in CGX post-processing mode.

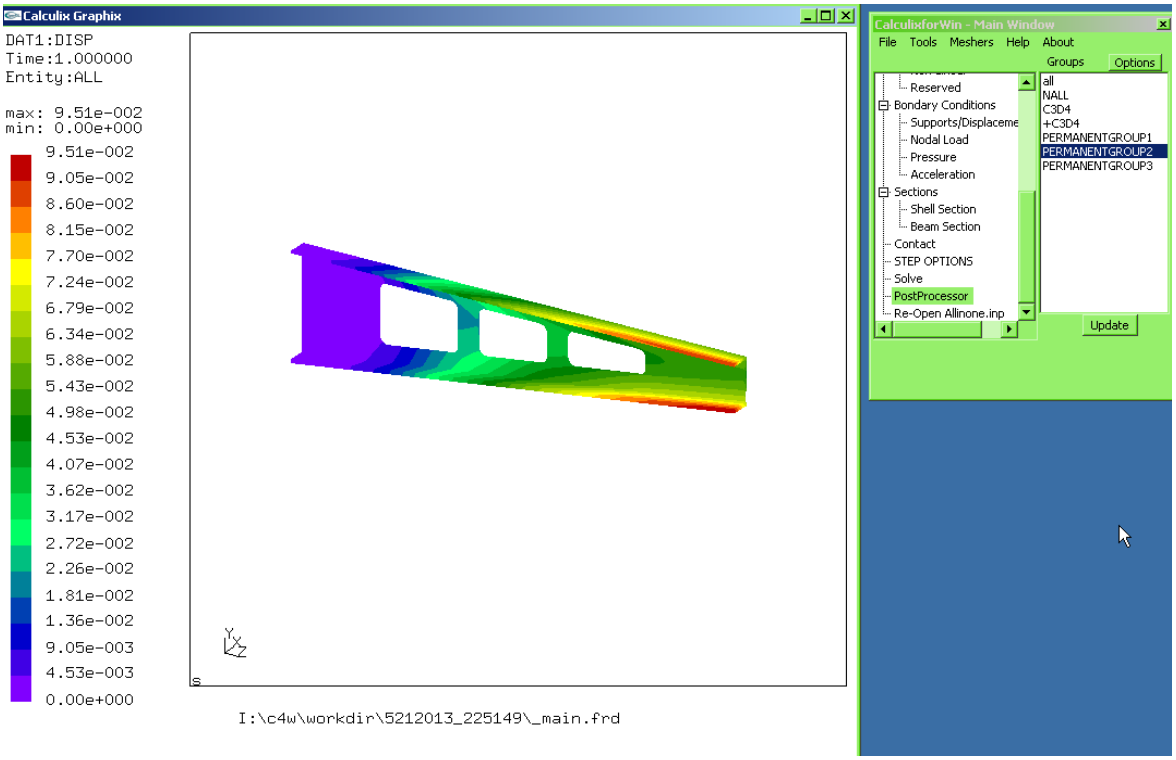


Fig. 30 - Show displacements/all. Maximal displacement is 0.0951 in.

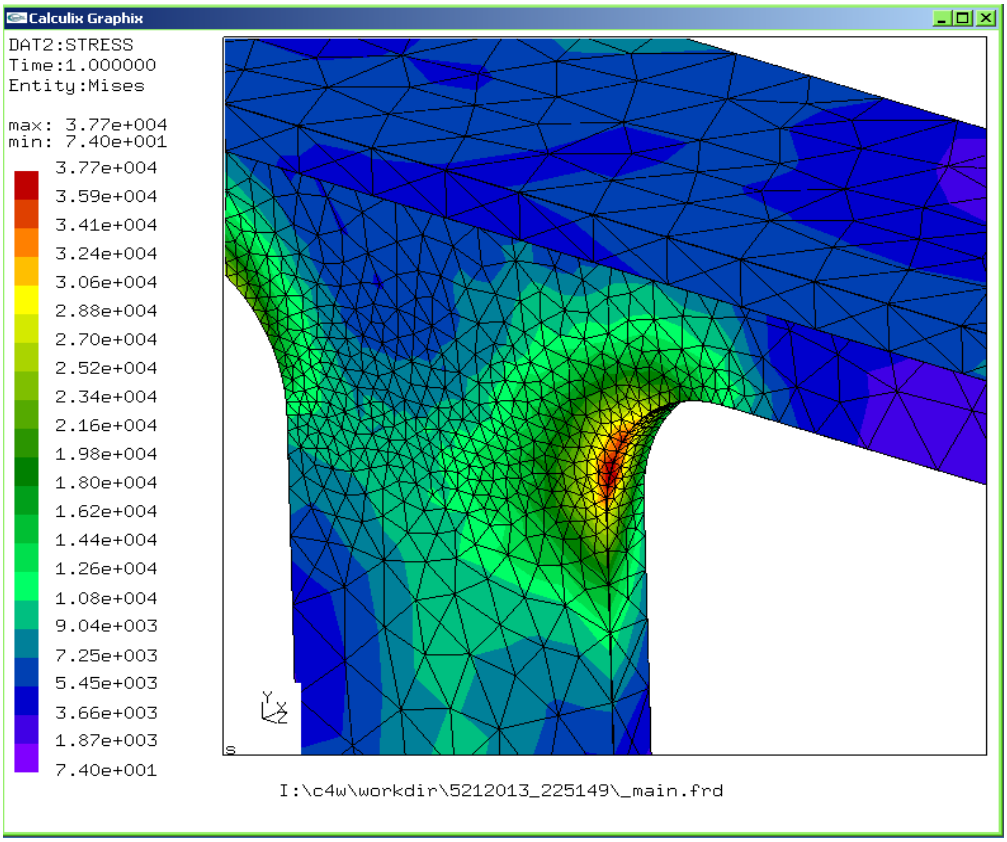


Fig. 31 Show Von-Mises stress. Max value is 37700 psi (in accordance with units were used).

Now we have nice looking pictures with results and everything looks fine except for the fact that the results are not correct! It is not correct due to fact that we were used C3D4 tetrahedral element first order elements without mid-nodes. These elements give slow convergence and make structure more rigid that it is. It's why this type of elements were implemented only in last versions of CalculiX (at the request of users).

Never use it for calculation!!!

Remember that nice looking pictures (in any program even it is not free and expensive) doesn't means proper results! Use verification examples every time when you want to learn new type of analysis.

Open GMSH .msh file (see note for Fig. 14), go to mesh>Set Order>"2" and save into the INP again. Repeat all steps above and solve the problem with quadratic elements. The correct results are shown below.

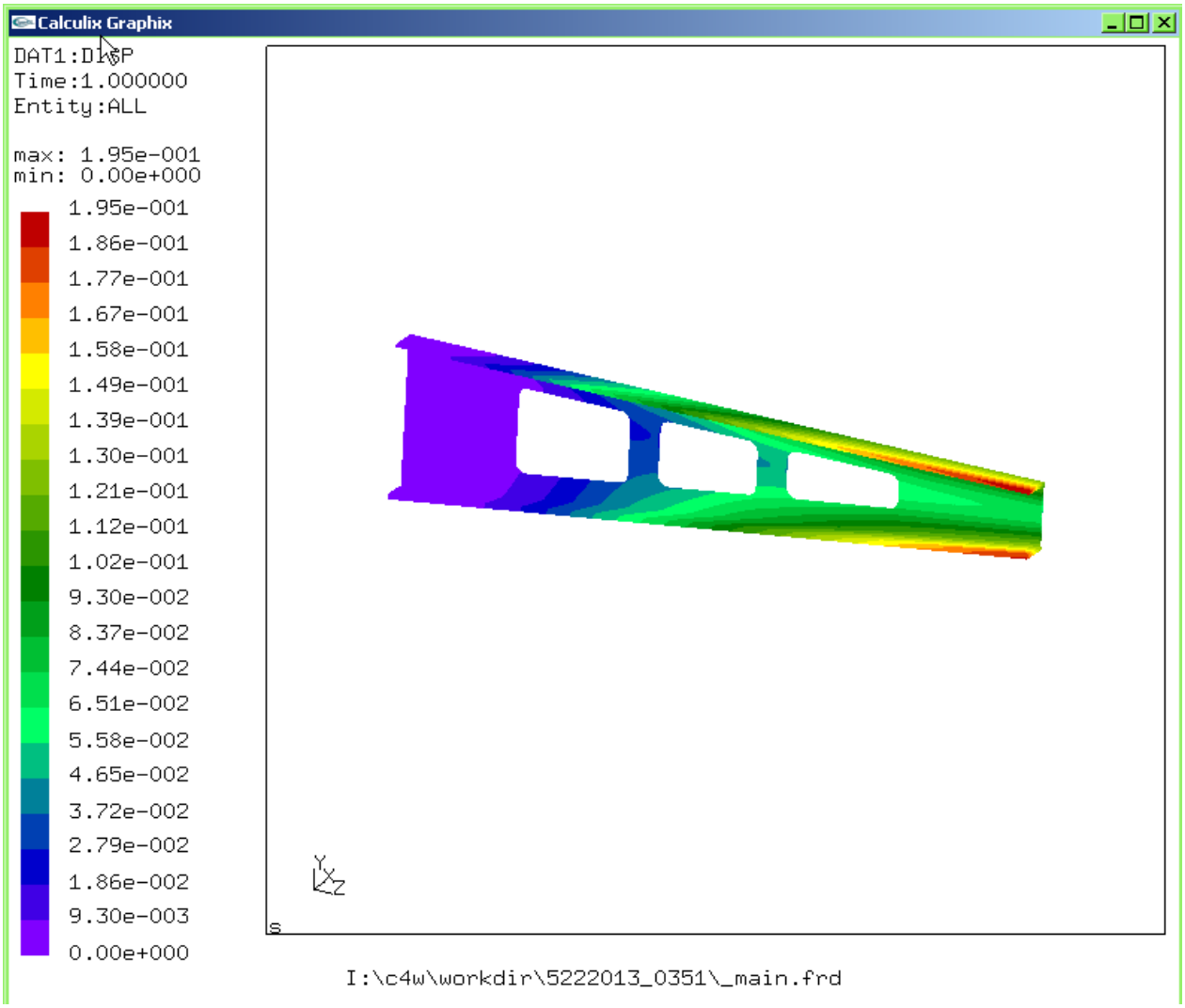


Fig. 32 - Displacements (max. 0.195 '')

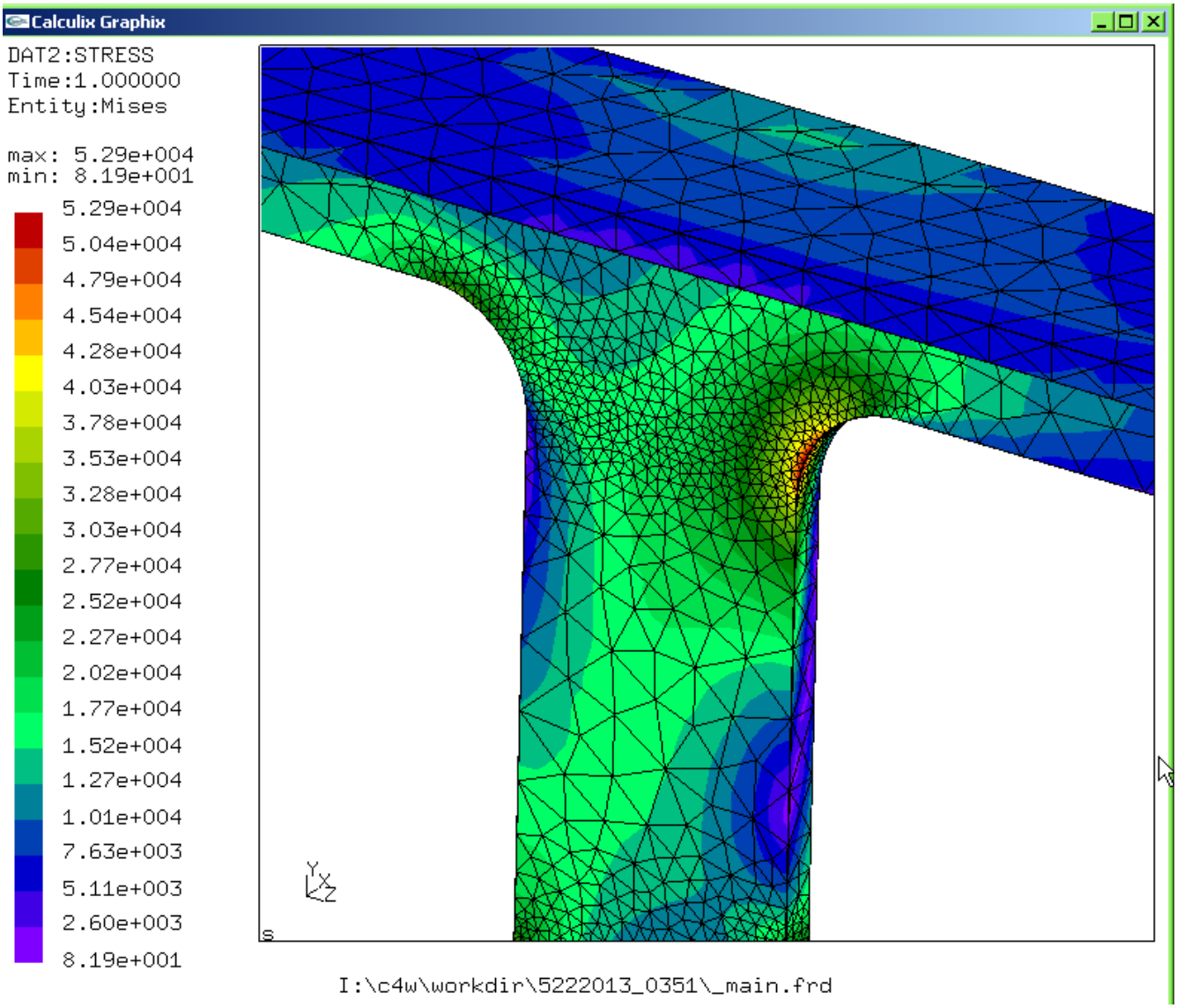


Fig. 33 - Von Mises stress (max. 52900 psi)

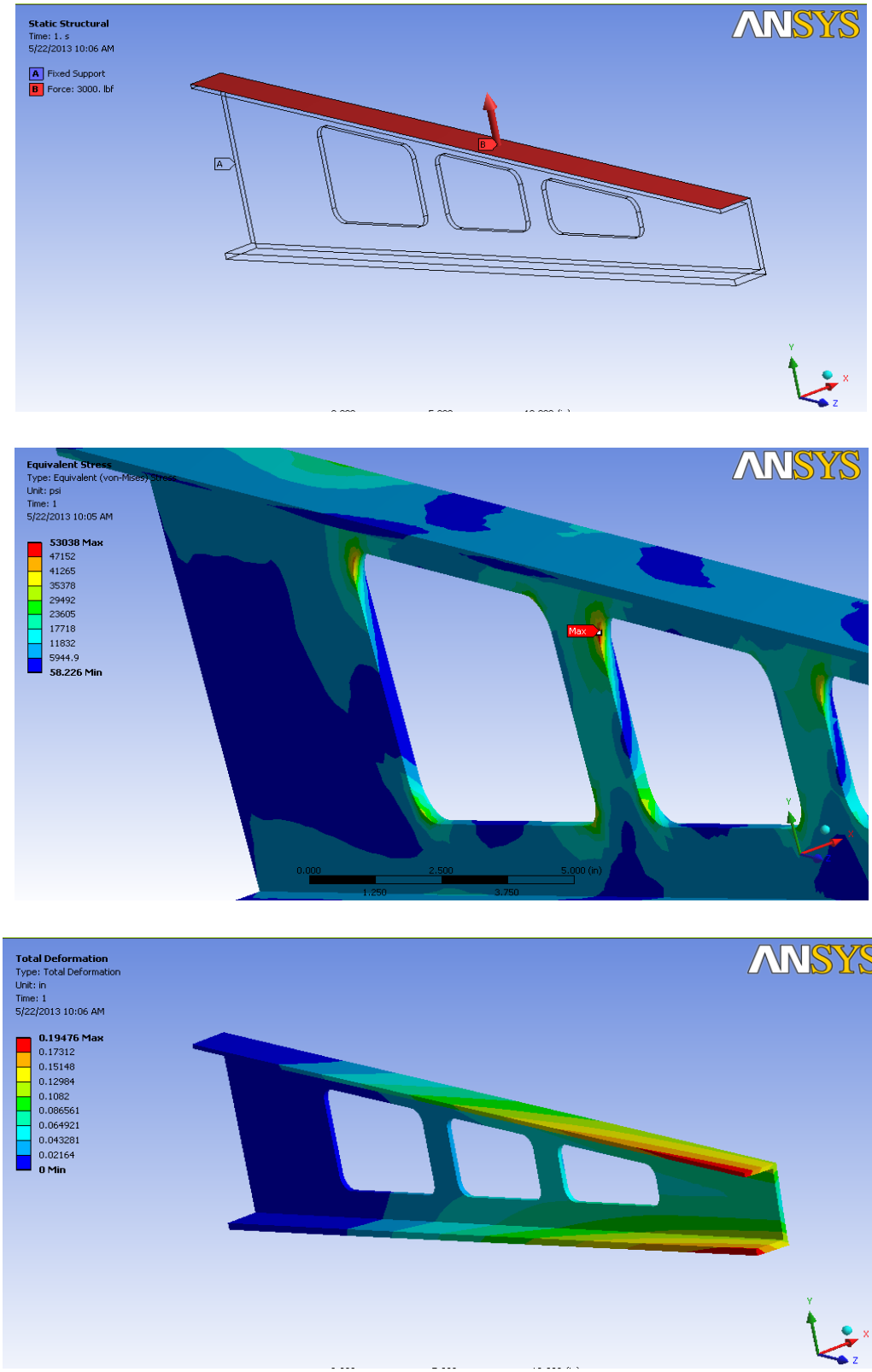


Fig. 34 - ANSYS Solution (Max. Displacement 0.195'', stress 53038 psi for current meshing of the model)

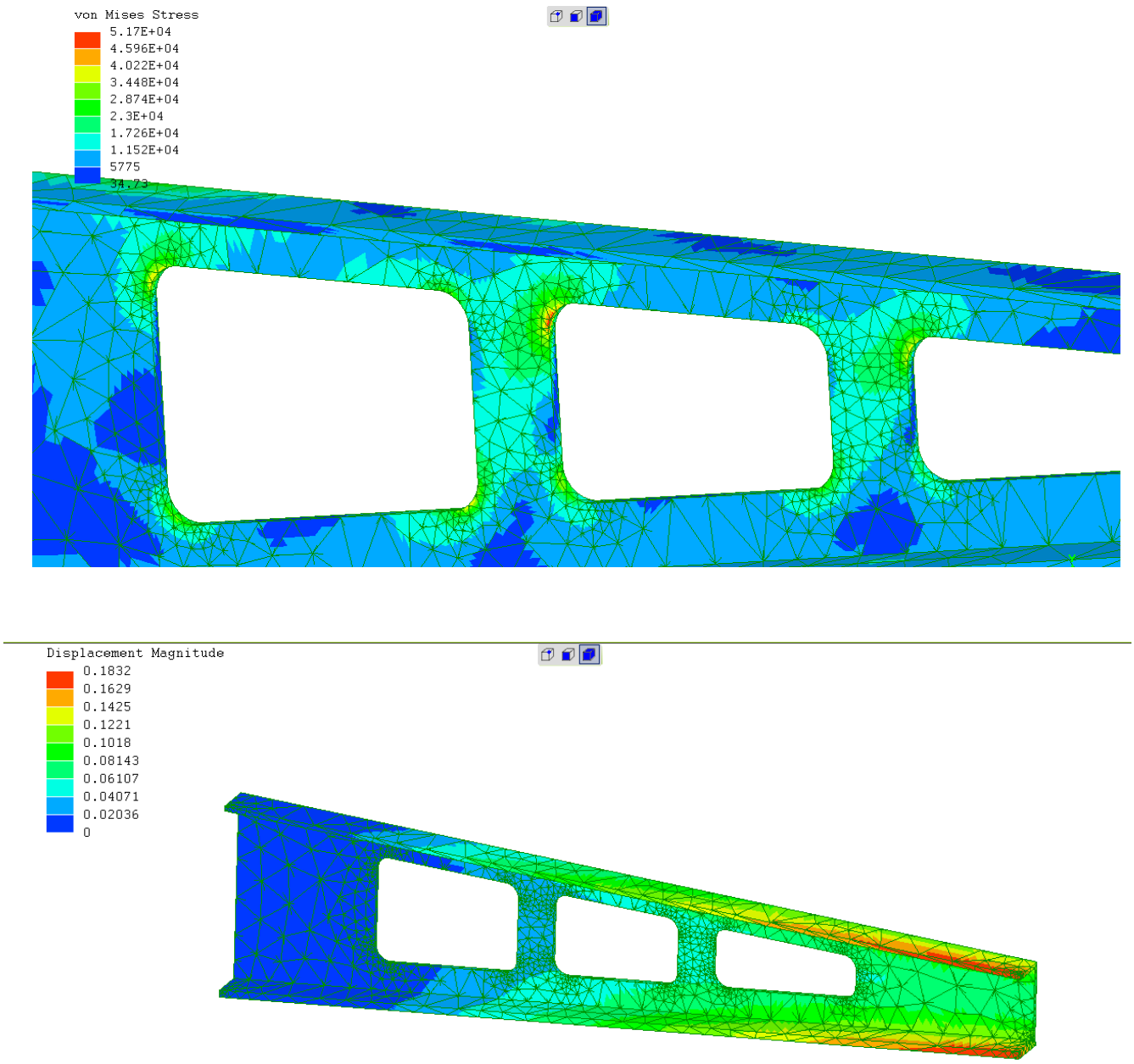


Fig. 35 Results obtained in Lisa-FET software
(Max. Displacement 0.183'', stress 51700 psi for current meshing of the model)